

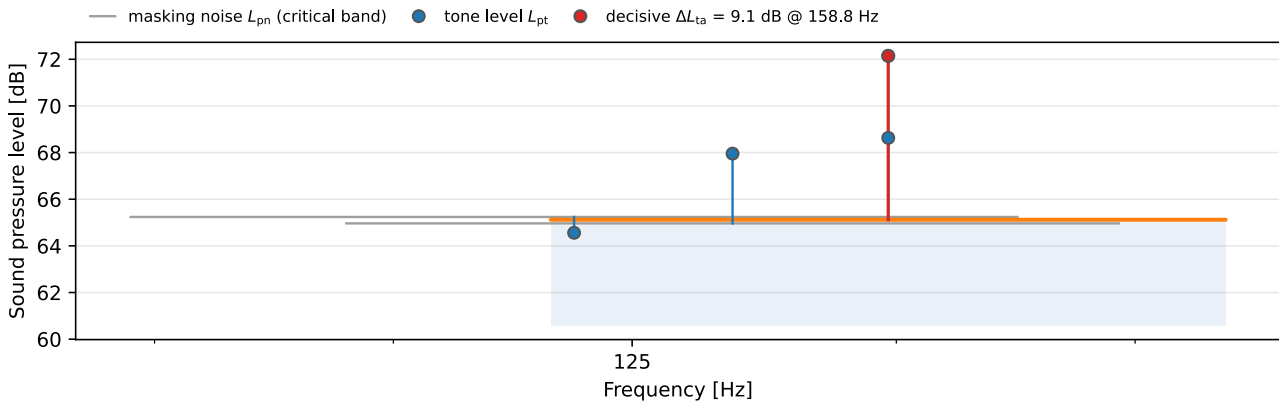
# Tonal audibility assessment

ISO 1996-2 tonal audibility by the ISO 1996-2:2017 Annex J engineering method (ISO/PAS 20065:2016); tonal adjustment K per Table J.1.

|                       |                                       |                                        |                                                       |
|-----------------------|---------------------------------------|----------------------------------------|-------------------------------------------------------|
| Source / situation:   | Combustion engine, steady operation   | Client:                                | Example client                                        |
| Measurement position: | Free field, 3 m from source (example) | Instrumentation:                       | Class 1 analyser, FFT with 2.7 Hz lines (Hann window) |
| Date of test:         | 2026-07-21                            | Analysis line spacing $\Delta f$ [Hz]: | 2.7                                                   |

## Detected tones

| Tone $f_T$ [Hz] | Type   | $L_{pt}$ [dB] | $L_{pn}$ [dB] | $\Delta f_c$ [Hz] | $\Delta L_{ta}$ [dB] | Present   |
|-----------------|--------|---------------|---------------|-------------------|----------------------|-----------|
| 118.4           | Single | 64.6          | 65.2          | 101               | 1.3                  | ● Present |
| 137.3           | Single | 68.0          | 65.0          | 101               | 5.0                  | ● Present |
| 158.8           | Single | 68.6          | 65.1          | 102               | 5.5                  | ● Present |
| 158.8           | FG (3) | 72.1          | 65.1          | 102               | 9.1                  | ● Present |



Tonal audibility  $\Delta L_{ta} = 9.1$  dB    tonal adjustment  
**K = 5 dB**

Decisive tone  $f_T = 158.8$  Hz  
Analysis line spacing  $\Delta f = 2.7$  Hz  
Extended uncertainty  $U = 3.3$  dB (90 % coverage)

Result vs requirement:  $\Delta L_{ta} = 9.1$  dB, required  $\leq 6.0$  dB → **FAIL**

A prominent tone is present (decisive  $\Delta L_{ta} = 9.1$  dB  $> 0$ ); the tonal adjustment  $K = 5$  dB (ISO 1996-2:2017 Table J.1) applies.

The tonal audibility  $\Delta L_{ta} = L_{pt} - L_{pn} - a_v$  is the amount by which the tone level rises above the masking threshold of the surrounding noise (ISO/PAS 20065:2016, Formula (14)); the decisive value is the largest over the detected tones (Clause 5.3.8).

$L_{pn}$  is the critical-band masking-noise level (Formula (12)),  $\Delta f_c$  the critical bandwidth about the tone (Formula (2)); an FG entry combines tones sharing a critical band (Formula (17)).

Table J.1 assigns K from the mean audibility  $\Delta L$ , the energy mean of the decisive audibilities of the J assessed spectra (ISO/PAS 20065:2016, Clause 5.3.9, Formula (20)). This fiche assesses a single spectrum, so the K shown follows from its decisive audibility; where several staggered spectra are assessed, rate their mean audibility instead.

**Laboratory:** Phonometry reference example

**Report no.:** EXAMPLE-1996-TONE

**Date:** 2026-07-21

Operator: phonometry    Signature: \_\_\_\_\_

■ The results relate only to the tested specimen.

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